

Planktons Diversity in Fresh Water Bodies Impacted By Sudden Discharge of Large Scale Fly Ash Slurry Due to Ash Pond Breach in IB Thermal Power Station At Banaharpali of Odisha, India

Malaya Ranjan Mahananda¹, Chander Kumar Singh²,
Arabinda Mishra³, Pramod Chandra Mishra⁴

¹Head, Post Graduate Department of Environmental Sciences, Sambalpur University, Jyoti Vihar – 768019, Sambalpur, Odisha

²Dept. of Environmental Sciences, Gautam Buddha University, Noida, Uttar Pradesh, India

³Chairman, Development and Environment Futures Trust, N1/303, IRC Village, Bhubaneswar-751015,

⁴Former Professor and Head, Dept. of Environmental Sciences, Sambalpur University, Res: Z1 Apartments, Tower 8, Flat 503, Nandankanan Road, Bhubaneswar-751024, Odisha

ABSTRACT

This investigation on assessing the impact of fly-ash slurry discharge due to Ash Pond breach reveals pronounced spatial and seasonal dynamics in the abundance, structure, and diversity of plankton communities in freshwater bodies located near Ib Thermal Power Station (ITPS). The transition from pre-monsoon to post-monsoon conditions served as a key environmental driver, eliciting clear ecological responses across both phytoplankton and zooplankton assemblages. Post-monsoon conditions favored an increase in total phytoplankton abundance at both the control and experimental sites. Taxonomic shifts were characterized by an expansion of Chlorophyceae and Bacillariophyceae, alongside a sharp decline in Cyanophyceae representation. While phytoplankton Shannon diversity remained stable at the control site, it exhibited a notable post-monsoon increase at the experimental site. The corresponding reduction in Simpson's dominance index reflects reduced single-taxon dominance, whereas the concurrent decline in species evenness highlights a less uniform distribution of individuals across the recorded taxa. Seasonal restructuring was even more pronounced within the zooplankton community. Pre-monsoon assemblages were primarily characterized by Copepoda and Rhizopoda, whereas Cladocera emerged as the dominant taxon post-monsoon. Although post-monsoon Shannon diversity increased across both study sites and Simpson's dominance decreased, the simultaneous drop in evenness indicates that overall diversity gains stemmed from shifting taxon representation rather than balanced numerical distribution. The study was carried out after almost one year of ash pond breach. Therefore, no conclusive evidence could be obtained on its negative impact on aquatic planktonic community. Two-way ANOVA test indicates that these variations were statistically significant. Both site and season independently influenced phytoplankton ($p < 0.001$) and zooplankton ($p < 0.001$ for site; $p = 0.020$ for season) abundances. Crucially, the interaction

between site and season was not significant for both phytoplankton ($p = 0.249$) and zooplankton ($p = 0.505$), establishing that seasonal shifts operated independently of location-specific differences.

Keywords: Phytoplankton , Zooplankton diversity ,dominance ,abundance ,Hirakud Reservoir Fly Ash slurry

INTRODUCTION

Thermal Power Plants are mega projects that require huge quantity of coal. Consequently, they have negative environmental impacts and put pressure on the local ecosystem. Ash sludge and fly ash represent a significant proportion of the solid waste arising from the combustion of coal in these plants. The soil and water bodies of the neighboring areas are physically and chemically contaminated by fly ash (Kakde & Sitre (2015). Fresh water is becoming a global problem as the rate of deterioration in water quality is increasing every day due to Municipality and Industrial waste water contamination (Mahananda et al. 2005). The aquatic organisms which include phytoplankton, zooplankton, benthos, fish and macrophytes are significantly impacted due to such contamination resulting in reduced aquatic productivity. Release of Coal fly ash in to water adversely affects the aquatic system's chemical composition and physical and biological components (Borm, 1997, Manz et al. 1999, Adriano et al, 1980, Ghio et al 2002, Saha et al. 2020). Rankhamb, & Mulgir, (2024) evaluated planktonic diversity as Bioindicators of environmental stress thermal/industrial discharge conditions. They demonstrated how changes in physicochemical parameters (turbidity, pH, trace metal loading) alter the composition, abundance, and dominance patterns of zooplankton species (Rotifera, Cladocera, Copepoda). Jaunjal and Gaupale (2025) in their review article on "Impact of Coal Fly Ash on Freshwater Fishes and the Ecosystem" opined that excess accumulation of fly ash in the water ecosystem significantly affects organisms by altering the physical properties of ecosystem, physiological biomechanisms, toxicity stress, and health issues. Consumption of contaminated foods results in the bioaccumulation and biotransformation of hazardous elements in the food chain that threaten to aquatic biodiversity.

The leaching of metals in aquatic systems leads to bioaccumulation, induction of oxidative stress, tissue damage, and alteration in blood biochemistry has been emphasized in the fishes. Excess accumulation of fly ash in the water ecosystem significantly affects organisms by altering the physical properties of ecosystem, physiological biomechanisms, toxicity stress, and health issues. Consumption of contaminated foods results in the bioaccumulation and biotransformation of hazardous elements in the food chain that threaten to aquatic biodiversity. Therefore, the present review aims to overview and highlight the toxicological impact of fly ash pollution on aquatic ecosystems and fish health. In addition, it is helpful to understand the level of toxicity and assess its effect on the environment.

The breach of ash pond/dyke C at the IB Thermal Power Station (ITPS) in Banharpalli of Jharsuguda district, Odisha, on December 9, 2023, led to the spillage of fly ash and slurry into surrounding agricultural fields, grazing lands, and water bodies. Fly ash, a by-product of coal combustion, contains trace elements and heavy metals such as arsenic (As), cadmium (Cd), lead (Pb), mercury (Hg), and iron (Fe), which pose significant environmental and health risks (Spencer et. al. 1983, Walia and Mehra, 1998, Yi et al. 2024). As planktons are the most important components of the aquatic ecosystem to maintain its stability and to adapt to any environmental changes, observation of the structure of phytoplankton communities can be a reliable method for biological monitoring studies to assess the state of pollution of aquatic organisms.

(Hossain et al. 2017). Thus, plankton population is used as a reliable tool for biomonitoring studies to assess the pollution status of aquatic bodies.

STUDY AREA

The present study sites are located in and around the Ib Thermal Power Station Power . The seven sampling locations in Banharpali represent the Experimental Sites whereas another seven in Rampela, far away from experimental locations served as control locations. The GPS coordinates of the selected sampling stations are presented in Table 1.

SAMPLING SITE	LATITUDE	LONGITUDE
Experimental 1	21.67723N	83.90495E
Experimental 2	21.67696N	83.90437E
Experimental 3	21.67916N	83.90221E
Experimental 4	21.67742N	83.89874E
Experimental 5	21.6786N	83.90204E
Experimental 6	21.69358N	83.90101E
Experimental 7	21.69355N	83.90058E
Control 1	21.65417N	83.95731E
Control 2	21.65508N	83.9598E
Control 3	21.65364N	83.96089E
Control 4	21.65303N	83.96323E
Control 5	21.65344N	83.9607E
Control 6	21.65362N	83.95932E
Control 7	21.70161N	83.88487E

Table 1: GPS Coordinates of the sampling sites

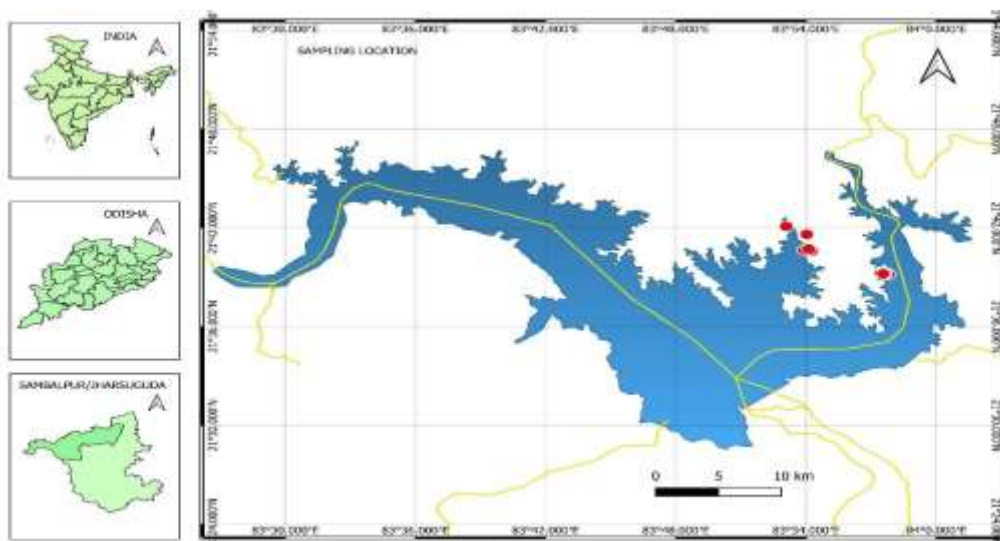


Fig 1: Map showing the locations of the study sites

MATERIALS AND METHODS

The collection of water samples were made during in pre-monsoon (May-June), and post-monsoon (October- November) between 9 to 10 am. The area was not accessible during monsoon due to spread-over water from the Hirakud Reservoir. For collection of samples, 14 stations (7 stations from each from Experimental and Control sites.

Plankton samples were collected by means of a bucket (50 litre) and filtered through plankton net of 50 μ . The filtrate was transferred to 50ml plastic bottle and preserved immediately in 4% formalin. The sample was thereafter taken to the laboratory for qualitative and quantitative analysis. Supernatant plankton was removed from free water and sedimentary plankton was counted by Sedgewick rafter counting cell (1 ml capacity) and then the planktons were identified under compound microscope.

To record the seasonal nature of plankton diversity in these location Shannon Diversity Index, Simpson Dominance Index (D), Evenness Index, Similarity Index (S) were used. These are the most suitable statistical tools to determine the phytoplankton species diversity

Shannon Diversity Index $H = \sum - (P_i * \ln P_i)$

Where, H = the Shannon diversity index P_i = fraction of the entire population made up of species S = numbers of species encountered $\sum$ = sum from species 1 to species S (Hossain et al.2017).)

Simpson Dominance Index (D) $H = \sum (n_i/n)^2$

Where, n_i = number of individuals of species, n=total number of individuals ((Hossain et al.2017)

Evenness Index $J = H/\ln S$

Where, J= Evenness index H=Shannon index S= max is the maximum possible value of H and it is equivalent to $\ln S$ (Hossain et al.2017)

RESULTS AND DISCUSSION

Pre-Monsoon							
Species	Control 1	Control 2	Control 3	Control 4	Control 5	Control 6	Control 7
Chlorophyceae							
Actinastrum sp.	+	+	-	-	+	-	+
Chlorella sp.	+	+	+	+	-	-	-
Chlamydomonas sp.	-	-	-	+	+	+	-
Cladophora sp.	+	+	+	+	+	+	+
Closterium sp.	+	+	+	+	+	-	+
Cosmarium sp.	-	+	+	+	+	+	+
Cylindrocapsa sp.	+	-	-	+	+	-	+
Desmidium sp.	+	+	-	+	+	+	+
Golenkinia sp.	+	-	-	+	-	-	-
Mougeotia sp.	+	+	+	+	+	+	+
Scenedesmus sp.	-	+	-	-	+	-	-

Sphaerosoma sp.	+	+	+	+	+	+	+
Spirogyra sp.	+	+	+	+	+	+	+
Triploceras sp.	+	-	+	-	-	+	-
Zygnema sp.	+	+	+	+	+	+	+
Cyanophyceae							
Anabaena sp.	+	+	+	+	+	+	+
Aulosira fertilissima	-	+	-	-	-	+	-
Chlorococcus sp.	-	-	+	+	-	-	+
Microcoleus	-	+	-	-	+	-	+
Lyngbya sp.	+	+	+	+	+	+	+
Nostoc sp.	+	+	+	+	+	+	+
Oscillatoria sp.	-	-	+	-	+	-	-
Rivularia sp.	+	-	-	+	-	-	+
Scytonema sp.	-	-	+	-	+	-	+
Spirulina sp.	+	-	+	-	+	-	+
Aphanizomenon	-	-	-	-	+	+	-
Gloeocapsa	+	+	+	+	+	+	+
Bacillariophyceae							
Cymbella sp.	+	+	+	+	-	+	+
Fragillaria sp.	+	+	-	-	+	+	+
Gyrosigma sp.	+	+	+	+	-	-	-
Navicula sp.	+	+	+	+	-	-	-
Nitzschia sp.	-	+	+	-	+	+	-
Synedra sp.	+	+	+	+	+	+	+
Euglenophyceae							
Euglena sp.	+	+	+	+	+	+	+
Chrysophyceae							
Diatom sp.	+	+	+	+	+	+	+
Vaculariaceae							
Gonyostomum sp.	+	-	-	+	-	-	-

Table 2: Presence phytoplankton in each station of the control sites during Pre-Monsoon

Post- Monsoon

Table 3: Presence phytoplankton in each station of the control sites

Species	Control 1	Control 2	Control 3	Control 4	Control 5	Control 6	Control 7
Chlorophyceae							
<i>Pediastrum sp.</i>	+	+	-	+	+	+	-
<i>Eurodina sp.</i>	-	+	+	-	+	+	-
<i>Cylindrocapsa sp.</i>	+	-	+	+	-	+	-
<i>Spirogyra sp.</i>	+	+	-	+	-	+	+
<i>Zygnema sp.</i>	+	+	+	+	+	+	+
<i>Golenkinia sp.</i>	-	+	-	+	-	-	-
<i>Desmidium sp.</i>	-	+	+	-	+	-	+
<i>Mougeotia sp.</i>	-	+	-	+	-	+	+
<i>Oscillatoria sp.</i>	+	+	+	+	+	+	+
<i>Cosmarium sp.</i>	-	+	+	-	+	-	-
<i>Tetraedron sp.</i>	-	-	-	+	-	+	-
<i>Cladophora sp.</i>	+	+	+	-	-	+	+
<i>Closterium sp.</i>	+	+	+	+	+	+	+
Bacillariophyceae							
<i>Cymbella sp.</i>	+	+	-	+	+	+	+
<i>Naviculla sp.</i>	+	-	+	+	+	-	+
<i>Leptocylindrus sp.</i>	+	+	+	+	+	+	+
<i>Nitzschia sp.</i>	+	+	+	-	+	+	+
<i>Fragillaria sp.</i>	-	+	+	+	+	-	+
<i>Synedra sp.</i>	+	+	+	+	+	+	+
<i>Bacillaria sp.</i>	+	-	+	-	+	+	+
<i>Coscinodiscaceae sp.</i>	-	+	-	+	-	+	+
<i>Gyrosigma sp.</i>	+	+	+	+	+	+	+
Cyanophyceae							
<i>Nostoc sp.</i>	+	+	+	+	+	+	+
<i>Chlorococcus sp.</i>	+	+	+	-	-	+	
<i>Merismopedia sp.</i>	-	+	-	+	+	+	+
<i>Gloeocapsa sp.</i>	+	+	-	-	-	-	+
Euglenaceae							
<i>Euglena sp.</i>	+	+	-	+	+	+	+
Chrysophyceae							
<i>Diatom sp.</i>	+	+	+	+	+	+	+

Table 3: Presence phytoplankton in each station of the control sites during Post-Monsoon

The tables 2 and 3 reveal similarities and differences in phytoplankton composition across the control locations in pre-monsoon and post-monsoon. Few species similar to all the seasons were *Cladophora*, *Closterium*, *Cosmarium*, *Cylindrocapsa*, *Desmidium*, *Golenkinia*, *Mougeotia*, *Spirogyra*, *Zygnema*, *Nostoc*, *Chlorococcus*, *Cymbella*, *Fragillaria*, *Nitzschia*, *Synedra*, *Euglena*, *Diatom* sp. Species including *Zygnema*, *Cladophora*, *Closterium*, *Nostoc*, *Synedra*, *Euglena* and *Diatom* sp. were reasonably widely distributed across the stations. However other species were confined to one season. Pre-monsoon had species like *Actinastrum*, *Chlorella*, *Chlamydomonas*, *Scenedesmus*, *Sphaerosoma*, *Triploceras*, *Anabaena*, *Aulosira*, *Microcoleus*, *Lyngbya*, *Rivularia*, *Scytonema*, *Spirulina*, *Aphanizomenon* and *Gonyostomum* while post-monsoon had species like *Pediastrum*, *Eurodina*, *Oscillatoria*, *Tetraedron*, *Bacillaria*, *Coscinodiscaceae*, *Merismopedia* and *Gloeocapsa*. To summarize, the two seasons had a good number of common species. However, the presence and distribution of several species were different in the two seasons, demonstrating seasonal fluctuation in the phytoplankton community.

Pre Monsoon							
Species	Exp. 1	Exp. 2	Exp. 3	Exp. 4	Exp. 5	Exp.6	Exp. 7
Chlorophyceae							
<i>Actinastrum</i> sp.	-	+	+	-	-	-	-
<i>Chlorella</i> sp.	-	+	-	+	-	-	-
<i>Chlamydomonas</i> sp.	-	-	-	+	+	+	-
<i>Cladophora</i> sp.	+	+	+	+	-	+	+
<i>Closterium</i> sp.	+	-	-	-	-	-	-
<i>Cosmarium</i> sp.	-	-	+	+	+	-	+
<i>Cylindrocapsa</i> sp.	+	-	+	+	+	+	+
<i>Desmidium</i> sp.	+	+	+	+	+	+	+
<i>Golenkinia</i> sp.	+	-	-	+	-	-	-
<i>Mougeotia</i> sp.	+	+	+	+	+	+	+
<i>Scenedesmus</i> sp.	-	+	-	-	-	-	-
<i>Sphaerosoma</i> sp.	+	+	+	+	+	+	+
<i>Spirogyra</i> sp.	+	+	+	+	+	+	+
<i>Triploceras</i> sp.	+	-	+	-	-	-	-
<i>Zygnema</i> sp.	+	+	+	+	+	+	+
Cyanophyceae							
<i>Anabaena</i> sp.	+	+	+	+	+	+	+
<i>Aulosira fertilissima</i>	-	+	-	-	-	-	-
<i>Chlorococcus</i> sp.	-	-	+	+	-	-	-

Microcoleus acutissimus	-	-	+	-	+	-	+
Lyngbya sp.	+	+	+	+	+	+	+
Nostoc sp.	+	+	+	+	+	+	+
Oscillatoria sp.	-	-	+	-	+	-	-
Rivularia sp.	-	-	+	-	-	-	-
Scytonema sp.	-	-	-	-	+	-	+
Spirulina sp.	+	-	+	-	-	-	-
Gloecapsa	+	+	+	+	+	+	+
Bacillariophyceae							
Cymbella sp.	+	+	+	+	-	-	+
Fragillaria sp.	+	+	-	-	+	+	+
Gyrosigma sp.	+	-	-	-	-	-	-
Navicula sp.	+	+	+	+	-	-	-
Nitzschia sp.	-	+	+	-	+	+	-
Synedra sp.	+	+	+	+	+	+	+
Euglenophyceae							
Euglena sp.	+	+	+	+	+	+	+
Chrysophyceae							
Diatom sp.	+	+	+	+	+	+	+

Table 4: Presence phytoplankton in each station of the experimental sites during Pre-Monsoon

Post-Monsoon							
Species	Exp 1	Exp 2	Exp 3	Exp 4	Exp 5	Exp 6	Exp. 7
Chlorophyceae							
<i>Pediastrum sp.</i>	+	-	+	+	+	-	+
<i>Cylindrocapsa sp.</i>	+	+	+	-	-	-	-
<i>Tetraedron sp.</i>	-	+	-	+	-	-	-
<i>Eudorina sp.</i>	+	+	-	+	-	+	+
<i>Chlamydomonas sp.</i>	-	-	+	-	+	-	-
<i>Spirogyra sp.</i>	+	+	+	+	+	+	+
<i>Zygnema sp.</i>	-	+	-	-	+	+	+
<i>Desmidium sp.</i>	+	-	+	-	+	+	-
<i>Mougeotia sp.</i>	+	+	+	+	+	+	+
<i>Cosmarium sp.</i>	+	+	+	+	-	-	+
<i>Gonatozygon sp.</i>	+	+	+	+	+	+	+

<i>Gloeocystis sp.</i>	+	+	-	-	+	-	+
<i>Closterium sp.</i>	+	-	+	+	+	-	+
Bacillariophyceae							
<i>Cymbella sp.</i>	+	+	+	+	+	+	+
<i>Naviculla sp.</i>	-	+	+	-	+	+	+
<i>Synedra sp.</i>	+	+	+	+	+	+	+
<i>Guinadia sp.</i>	+	-	+	-	-	-	-
<i>Pinnularia sp.</i>	+	+	-	-	+	+	+
<i>Leptocylindrus sp.</i>	+	+	+	+	+	+	+
<i>Nitzschia sp.</i>	+	+	+	+	+	+	+
Cyanophyceae							
<i>Nostoc sp.</i>	+	+	+	+	+	+	+
<i>Chlorococcus sp.</i>	-	+	+	-	+	-	-
<i>Merismopedia sp.</i>	+		-	-	+	-	+
<i>Lyngbya sp.</i>	+	+	+	+	+	+	+
<i>Gloeocapsa sp.</i>	-	+	-	+	-	+	+
Euglenaceae							
<i>Euglena Sp.</i>	+	+	+	+	+	+	+

Table 5: Presence of phytoplankton in experimental sites during Monsoon

In the experimental sites between the two seasons there are some common species in the phytoplankton community but also with notable variations. Common species include *Cylindrocapsa*, *Chlamydomonas*, *Spirogyra*, *Zygnema*, *Desmidium*, *Mougeotia*, *Cosmarium*, *Closterium*, *Cymbella*, *Synedra*, *Nitzschia*, *Nostoc*, *Chlorococcus*, *Lyngbya*, *Gloeocapsa* and *Euglena*. The species *Spirogyra*, *Mougeotia*, *Cymbella*, *Synedra*, *Nitzschia*, *Nostoc*, *Lyngbya* and *Euglena* were generally dispersed in the samples. Pre-monsoon featured certain species that were not listed in the post-monsoon such as *Actinastrum*, *Chlorella*, *Golenkinia*, *Scenedesmus*, *Sphaerosoma*, *Triploceras*, *Anabaena*, *Aulosira*, *Microcoleus*, *Oscillatoria*, *Rivularia*, *Scytonema*, *Spirulina*, *Fragillaria*, *Gyrosigma* and *Navicula*. In the post-monsoon, however, there were a number of species which did not occur during pre-monsoon, e.g. *Pediastrum*, *Tetraedron*, *Eudorina*, *Gonatozygon*, *Gloeocystis*, *Guinadia*, *Pinnularia*, *Leptocylindrus* and *Merismopedia*.

	Chloro- phyceae	Cyano- phyceae	Bacillario- phyceae	Eugleno- phyceae	Chryso- phyceae	Vaculari- aceae	Total
Control 1	61	54	24	12	10	2	163
Control 2	50	39	23	9	8	0	129
Control 3	41	48	19	11	12	0	131
Control 4	60	32	18	10	14	3	137
Control 5	57	53	9	9	9	0	137

Control 6	35	41	15	4	12	0	107
Control 7	37	38	15	7	6	0	103
Average	48.71	43.57	17.57	8.86	10.14	0.71	129.57
Percent Value (%)	37.6	33.63	13.56	6.84	7.83	0.55	100%

Table 6: Relative abundance of Phytoplankton in pre- monsoon (Nos./ml) in control site

	Chloro-phyceae	Bacillario-phyceae	Cyano-phyceae	Euglenaceae	Chryso-phyceae	Total
Control 1	63	55	18	4	9	149
Control 2	74	49	21	6	7	157
Control 3	64	53	17	0	6	140
Control 4	66	54	16	8	5	149
Control 5	68	53	26	4	8	159
Control 6	57	46	27	3	2	135
Control 7	63	52	18	5	10	148
Average	65.00	51.71	20.43	4.29	6.71	148.14
Percent Value	43.88	34.91	13.79	2.89	4.53	100

	Chloro-phyceae	Bacillario-phyceae	Cyano-phyceae	Euglenaceae	Chryso-phyceae	Total
Experimental 1	30	41	13	8	7	99
Experimental 2	32	29	14	3	4	82
Experimental 3	27	45	10	4	10	96
Experimental 4	26	30	9	4	4	73
Experimental 5	22	33	14	3	7	79
Experimental 6	23	16	4	2	3	48
Experimental 7	21	17	4	2	2	46
Average	25.86	30.14	9.71	3.71	5.29	74.71
Percent Value	34.61	40.34	13	4.97	7.07	100

Table 7: Relative abundance of Phytoplankton in pre- monsoon (Nos./ml) in experimental site

	Chlorophyceae	Bacillariophyceae	Cyanophyceae	Euglenaceae	Total
Experimental 1	50	40	19	5	114
Experimental 2	47	38	25	4	114
Experimental 3	49	40	20	5	114
Experimental 4	43	28	24	2	97

Experimental 5	46	36	24	6	112
Experimental 6	36	40	21	3	100
Experimental 7	34	38	24	4	100
Average	43.57	37.14	22.43	4.14	107.29
Percent Value	40.61	34.62	20.91	3.86	100

Table 8: Relative abundance of Phytoplankton in post monsoon (Nos./ml) in experimental sites

	Pre-Monsoon		Post-Monsoon	
	Control	Experimental	Control	Experimental
Chlorophyceae	48.71	25.86	65	43.57
Cyanophyceae	43.57	30.14	20.43	22.43
Bacillariophyceae	17.57	9.71	51.71	37.14
Euglenophyceae	8.86	3.71	4.29	4.14
Chrysophyceae	10.14	5.29	6.71	0
Vaculariaceae	0.71	0	0	0

Table 9: Abundance of Phytoplankton in Nos/ml

The average phytoplankton composition across all seven sampling stations showed clear seasonal variation. During the pre-monsoon season, **Chlorophyceae** was dominant in both control (48.71%) and experimental (25.86%) sites, followed by **Cyanophyceae** (43.57% and 30.14%, respectively). In the post-monsoon season, Chlorophyceae increased to **65% in the control and 43.57% in the experimental site**, while **Bacillariophyceae** also increased markedly to 51.71% and 37.14%, respectively. In contrast, Cyanophyceae declined to 20.43% and 22.43%. Overall, the post-monsoon season was characterized by higher contributions of Chlorophyceae and Bacillariophyceae and a decline in Cyanophyceae.

	Pre-Monsoon		Post-Monsoon	
	Control	Experimental	Control	Experimental
Chlorophyceae	37.60%	34.61%	43.88%	40.61%
Cyanophyceae	33.62%	40.34%	13.79%	20.91%
Bacillariophyceae	13.56%	13%	34.91%	34.62%
Euglenophyceae	6.84%	4.97%	2.89%	3.86%
Chrysophyceae	7.83%	7.07%	4.53%	0.00%
Vaculariaceae	0.55%	0.00%	0.00%	0.00%

Table 10: Percent Value in Abundance of Phytoplankton

The table 10 shows the percentage of phytoplankton classes that were dominant at the control and experimental sites during the pre- and post-monsoon period. Chlorophyceae contributed 34.61 percent to

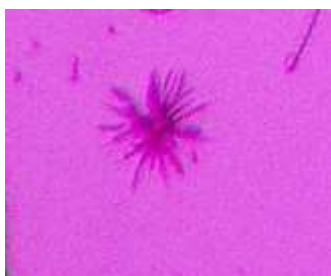
the experimental site. Bacillariophyceae represented 13.56 and 13.00 percent of the control and study populations, respectively. Euglenophyceae represented 6.84 percent of the control and 4.97 percent of the experimental population, while Chrysophyceae represented 7.83 percent and 7.07 percent. Vasculariaceae were only observed in the control site (0.55%).

In the post-monsoon period, chlorophyll became dominant in both locations, with 43.88 percent in the control setting and 40.61 percent in the experimental setting. Bacillariophyceae also increased significantly, by 34.91 percent in control and 34.62 percent in the experimental setting. Cyanophytes, on the other hand, decreased significantly to 13.79 percent in the control group and 20.91 percent in the experimental group. Euglenophyceae declined to 2.89 and 3.86 percent. Chrysophyceae decreased to 4.53 percent in the control group and was completely absent (0) in the experimental group. In both places, Vaculariaceae were absent during the post-monsoon period.

Phytoplankton communities showed a clear seasonal change in composition. The pre-monsoon period was characterized by a relatively higher presence of cyanophyceae, especially at experiment. Sites. On the other hand, the post-monsoon period showed an increase in chlorophyll and Bacillariophyceae at both the locations. Chlorophyte levels increased from 37.60 to 43.88 percent in control and from 34.61 to 40.61 percent in the experimental sites. The most significant seasonal change was observed in cyanophyceae, which decreased from 13.56 to 13.79 percent in control and from 40.34 to 20.62 percent in the experimental setting. Thus, the phytoplankton community after the monsoon is more strongly represented by the chlorophyceae and the Bacilliophyceae, while the cyanophyceae have a higher relative contribution during the pre-monsoon period.



Spirogyra sp.



Actinastrum sp.



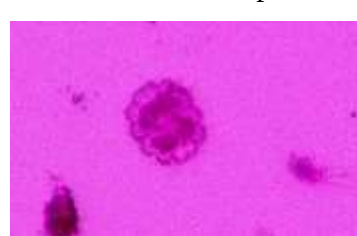
Chlorella sp.



Cladophora sp.



Closterium sp.



Cosmarium sp.



Cylindrocapsa sp.



Desmidium sp.



Mougeotia sp.



Scenedesmus sp.



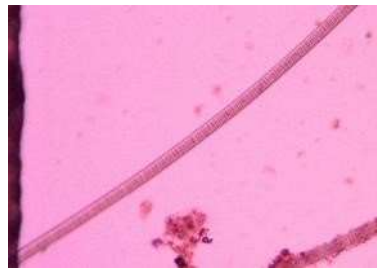
Sphaerosasma sp.



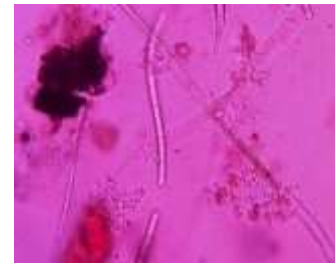
Zygnema sp.



Anabaena sp.



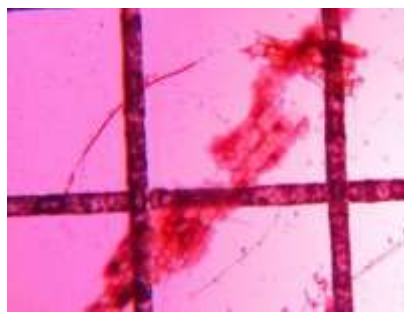
Microcoleus acutissimus



Nostoc sp



Rivularia sp.



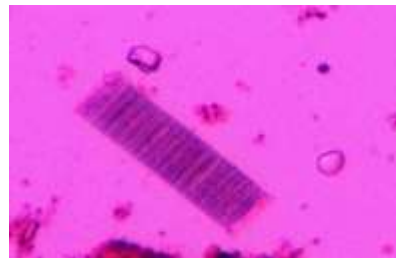
Aphanizomenon



Gloeocapsa



Cymbella sp.



Fragillaria sp.



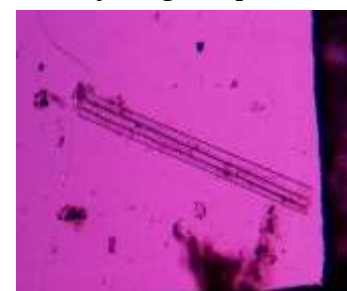
Gyrosigma sp.



Navicula sp.



Nitzschia sp.



Synedra sp.

Fig 2 Species of the phytoplankton recorded during study period

Species	Control 1	Control 2	Control 3	Control 4	Control 5	Control 6	Control 7
Rotifer							
Anuraeopsis sp.	-	-	+	+	-	-	+
Polyarthra sp.	-	+	-	+	+	-	+
ascomorpha	+	+					
Cladoceran							
Polyphemus sp.	+	-	-	-	-	-	-
Alona affinis sp.	-	+	-	-	-	+	-
Leptodora sp.	-	+	-	+	-	+	-
Diaphanosoma sp.	-	-	-	+	-	-	-
Monia sp.	-	-	-	+	-	-	
Bosmina sp.	-	-	-	-	-	-	+
Copepoda							
Diaptomus sp.	+	-	+	-	+	+	+
Phyllodiaptomus sp.	-	-	+	-	+	-	+
Paradipatomus sp.	+	-	+	-	+	+	-
Cyclops sp.	-	+	-	+	+	-	+
Rizopoda							
Paramecium sp.	+	+	+	+	+	+	+
Arcella sp.	-	-	+	+	-	-	-
Protozoa							
Volvox sp.	+	-	-	-	-	+	-

Table 11: Distribution of Zooplanktons Control site during Pre- Monsson

Species	Control 1	Control 2	Control 3	Control 4	Control 5	Control 6	Control 7
Cladoceran							
Diaphanosoma sp.	+	-	+	+	-	+	+
Bosmania sp.	-	+	+	-	+	-	+
Alona affins sp.	+		+	+	-	+	-
Monia sp.	+	+	-	-	-	+	-
Daphnia sp.	-	+	+	-	+	-	+
Polyphemus sp.	-		+	-	-	+	-

Copepoda							
Diaptomus sp.	-	+	-	-	-	-	+
Phyllodiaptomus sp.	+	-	+	-	+	+	-
Paradipatomus sp.	+	+	-	-	-	+	+
Cyclops sp.	-	-	+	-	+	-	+
Rhizopoda							
Paramecium sp.	+	+	+	+	+	+	+
Branchionide							
Karatella cochlearis sp.	+	+	-	+	-	+	+
Rotifer							
Monostyla sp.	+	+	+	+	-	+	+
Anuraeopsis sp.	-	-	+	-	-	+	-
Crustacea							
Nauplius sp.	+	+	-	+	+	-	+
Protozoa							
Volvox sp.	-	+	-	-	+	-	-

Table 12: Presence of Zooplanktons in control site during post- Monsoon

Between pre-monsoon and post monsoon few common species of zooplankton at the control site were recorded. Those are *Diaphanosoma*, *Bosmina/Bosmania*, *Alona affinis*, *Monia*, *Polyphemus*, *Diaptomus*, *Phyllodiaptomus*, *Paradipatomus*, *Cyclops*, *Paramecium* and *Volvox*. *Paramecium* was present at all seven stations in both seasons which is the most consistent distribution as observed. But some species were present only in one season. In the pre monsoon *Anuraeopsis*, *Polyarthra*, *Ascomorpha* and *Leptodora* and in post monsoon *Daphnia*, *Monostyla*, *Karatella cochlearis* and *Nauplius* were recorded. Also the abundance and station-wise distribution of species changed between the seasons; for example, *Diaphanosoma* and *Alona affinis* saw an increase in their wide spread distribution in the post monsoon. As a whole the results point to a moderate seasonal variation in zooplankton make up, with many of the same taxa present in both seasons.

Species	Exp. 1	Exp2	Exp. 3	Exp.4	Exp. 5	Exp. 6	Exp. 7
Rotifer							
Anuraeopsis sp.	-	-	+	+	-	-	+
Polyarthra sp.	-	+	-	+	+	-	+

Cladoceran							
Polyphemus sp.	+	-	-	-	-	-	-
Alona affinis sp.	-	+	-	-	-	+	-
Copepoda							
Diaptomus sp.	+	-	-	-	+	+	
Phyllodiaptomus sp.	-	-	+	-	-	-	+
Paradipatomus sp.	+		+	-	-	+	-
Cyclops sp.	-	+	-	+	-	-	+
Rizopoda							
Paramecium sp.	+	+	+	+	+	+	+

Table 13: Distribution of zooplanktons in experimental site during Pre -Monsoon

Species	Control 1	Control 2	Control 3	Control 4	Control 5	Control 6	Control 7
Cladoceran							
<i>Diaphanosoma sp.</i>	-	+	+	-	+	-	-
<i>Bosmania sp.</i>	+	-	-	+	-	-	+
<i>Alona affins sp.</i>	+	+	-	-	+	-	-
<i>Moina sp.</i>	-	+	+	-	-	+	-
<i>Daphnia sp.</i>	+	-	+	+	+	-	+
<i>Leptodora sp.</i>	-	-	+	-	-	+	-
Copepoda							
<i>Phyllodiaptomus sp.</i>	+	-	+	-	-	-	-
<i>Paradipatomus sp.</i>	-	-	+	-	+	-	+
<i>Cyclops sp.</i>	+	+	-	-	+	-	+
Rhizopoda							
<i>Paramecium sp.</i>	-	-	+	+	+	+	+
Crustacea							
<i>Nauplius sp.</i>	+	+	-	-	-	-	+
Protozoa							
<i>Volvox sp.</i>	-	+	+	-	+	+	-

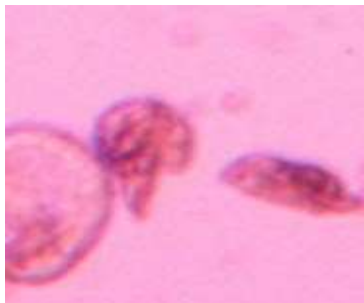
Table 14: Distribution of Zooplanktons Experimental site during Post- Monsoon

Nine species have been recorded in the pre-monsoon period, such as *Anuraeopsis*, *Polyarthra*, *Polyphemus*, *Alona affinis*, *Diatomus*, *Phyllodiaptomus*, *Paradipomus*, *Cyclops*, and *Paramecium*. In the post-monsoon period, eleven taxa were observed, including the new arrivals *Diaphanosoma*, *Bosmina*, *Moina*, *Daphnia*, *Leptodora* and *Nauplius*. Some pre-monsoon taxa, in particular *Anuraeopsis*, *Polyarthra*, *Polyphemus* and *Diaspora*, did not appear after the monsoon. *Paramecium* was widely distributed before the monsoon, which decreased after the monsoon. An increase was also observed in

zooplankton diversity, from 9 in pre-monsoon to 11 post-monsoons. Overall, the post-monsoon period showed a higher diversity of zooplankton (11 taxa) than the pre-monsoon period (9 taxa), suggesting seasonal changes in the zooplankton community in the experimental site.



Phyllodiaptomus sp.



Polyphemus sp.



Paradiaptomus sp.



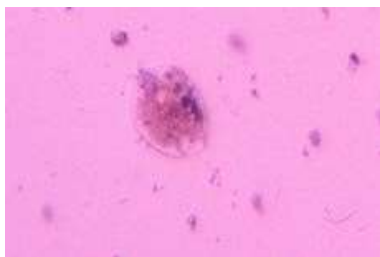
Alona affinis sp.



Cyclops sp.



Ascomorpha sp.



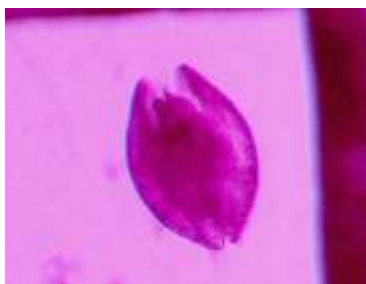
Monia sp.



Diaptomus sp.



Paramesium sp.



Anuraaeopsis sp.



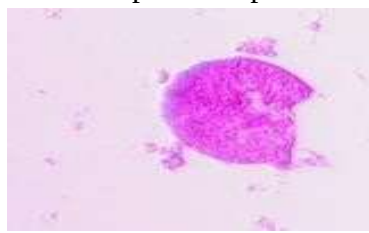
Leptodora sp.



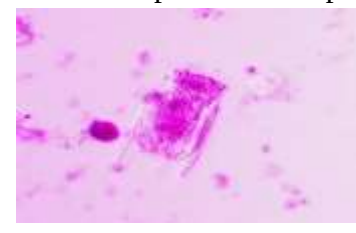
Diaphanosoma Sp.



Arcella sp.



Volvox sp.



Polyarthra sp.

Fig 3 Species of the Zooplanktons recorded during study period

	Rotifer	Cladoceran	Copepoda	Rizopoda	Protozoa	Total
Control 1	1	3	11	6	3	24
Control 2	7	8	3	4	0	22
Control 3	2	0	10	8	0	20
Control 4	7	7	4	12	0	30
Control 5	6	0	23	8	0	37
Control 6	0	11	10	9	4	34
Control 7	6	4	13	7	0	30
Average	4.14	4.71	10.57	7.71	1.00	28.14
Percent Value	14.72	16.75	37.56	27.41	3.55	100.00

Table 15: Relative abundance Zooplankton abundance in pre- monsoon (Nos./ml) in control site

	Cladoceran	Copepoda	Rhizopoda	Branchionide	Rotifer	Crustacea	Protozoa	Total
Control 1	13	12	5	0	7	0	0	37
Control 2	14	7	3	4	2	4	4	38
Control 3	14	8	4	3	3	0	0	32
Control 4	9	1	4	5	0	1	0	20
Control 5	9	7	3	0	5	2	3	29
Control 6	13	12	6	0	3	0	3	37
Control 7	12	7	5	5	0	3	0	32
Average	12.00	7.71	4.29	2.43	2.86	1.43	1.43	32.14
Percent Value	37.67	24.22	13.45	7.62	8.97	4.48	3.59	100

Table 16: Relative abundance of Zooplanktons in post- monsoon (Nos./ml) in Control sites

	Rotifer	Cladoceran	Copepoda	Rizopoda	Total
Experimental 1	0	2	3	6	11
Experimental 2	4	2	1	8	15
Experimental 3	2	0	5	3	10

Experimental 4	5	0	2	5	12
Experimental 5	1	0	1	4	6
Experimental 6	0	1	5	6	12
Experimental 7	7	0	6	3	16
Average	2.71	0.71	3.29	5.00	11.71
Percent Value (%)	23.17	6.10	28.05	42.68	100

Table 17: Relative abundance Zooplankton abundance in pre- monsoon (Nos./ml) in experimental site

Table 19: Relative abundance of Zooplanktons in post monsoon (Nos./ml) sample in Experimental sites

	Cladoceran	Copepoda	Rhizopoda	Crustacea	Protozoa	Total
Experimental 1	12	3	2	3	0	20
Experimental 2	7	1	4	2	5	19
Experimental 3	15	7	6	0	2	30
Experimental 4	5	0	7	0	0	12
Experimental 5	10	6	4	0	2	22
Experimental 6	5	0	0	0	4	9
Experimental 7	5	5	5	4	0	19
Average	8.43	3.14	4.00	1.29	1.86	18.71
%Value	45.04	16.79	21.37	6.87	9.92	100

Table 18: Relative abundance of Zooplanktons in post-monsoon (Nos./ml) sample in Experimental sites

	Pre-Monsoon		Post-Monsoon	
Taxa	Control	Experimental	Control	Experimental
Rotifer	4.14	2.71	2.86	0
Cladoceran	4.71	0.71	12	8.43
Copepoda	10.57	3.29	7.71	3.14
Rizopoda	7.71	5	4.29	4
Protozoa	1	0	1.43	1.86
Branchionide	0	0	2.43	0
Crustacea	0	0	1.43	1.29

Table 19: Abundance of Zooplanktons in Nos/ml

The average composition of zooplanktons at all seven sampling points showed a clear seasonal variation. During the pre-monsoon period, Copepoda was dominant in both control (10.57) and experimental (3.29)

areas, followed by Rizopoda (7.71 and 5.00 percent respectively). In the post-monsoon period, Cladoceran was the dominant group, increasing to 12.00 percent in the control group and 8.43 percent in the experimental group. Copepoda and Rizopoda declined, while Branchionid and Crustacea appeared after the monsoon. Overall, the pre-monsoon community was represented mostly by Copepods and Rizopods, while the post-monsoon community was more represented by Cladocerans, suggesting a distinct seasonal change in the composition of zooplankton.

Taxa	re-Monsoon		Post-Monsoon	
	Control	Experimental	Control	Experimental
Rotifer	14.72%	23.17%	8.97%	0.00%
Cladoceran	16.75%	6.10%	37.67%	45.04%
Copepoda	37.56%	28.05%	24.22%	16.79%
Rizopoda	27.41%	42.68%	13.45%	21.37%
Protozoa	3.55%	0.00%	3.59%	9.92%
Branchionide	0.00%	0.00%	7.62%	0.00%
Rotifer	0.00%	0.00%	0.00%	0.00%
Crustacea	0.00%	0.00%	4.48%	6.87%

Table 20: Percent Value in Abundance of Phytoplankton

Table 20 shows the percentage of the different zooplankton groups found in the control and experimental areas during the pre- and post-monsoon period. In the Pre-Monsoon predominant zooplankton group in the control area was Copepoda with 37.56 percent, followed by Rizopoda (27.41%), Cladoceran (16.75) and Rotifer (14.72). Protozoa contributed only 3.55 percent, and neither branchionide nor crustacea were present. The dominant group was Rizopoda (42.68%), followed by Copepoda (28.05) and Rotifer (23.17). Cladoceran contributed 6.10 percent, with no sign of the presence of Protozoa, Branchionide or Crustacea. The composition changed significantly during the post-monsoon period. The Cladoceran group was dominant in both settings, accounting for 37.67 percent of the control group and 45.04 percent of the experimental group. Copepoda decreased to 24.22 percent in control and 16.79 percent in the experimental setting, while Rizopoda contributed 13.45 and 21.37 percent to the total. Rotifer was 8.97 percent in the control group, but was not present in the experimental sites. Protozoa contributed 3.59 percent to the control and increased to 9.92 percent in the experimental samples. Branchionide was only detected in the control site (7.62%), while Crustacea contributed 4.48 and 6.87 percent, respectively, to the control and study sites.

A clear seasonal change in the composition of zooplankton has been observed between both seasons. During the pre-monsoon period, the community was predominantly composed of Copepoda and Rizopoda, with Copepoda dominating at the control site (37.56) and Rizopoda dominating at the experimental site (42.68). In contrast, after monsoon season, cladocera was dominant in both sites, increasing from 16.75 to 37.67 percent in the control and from 6.10 to 45.04 percent in the experimental samples. Rizopoda was also reduced in the control site but showed a relative increase in the experimental site in comparison to pre-monsoon period. Protozoa increased from 0 to 9.92 percent in the experimental site, while Branchionid

and Crustacea appeared after the monsoon. In the experimental site, Rotifer was significantly reduced, from 23.17 percent pre-monsoon to 0 in post-monsoon, with the appearance or increase of several other groups of zooplanktons. Overall, the pre-monsoon assemblage was dominated by Copepods and Rizopods, while the post-monsoon assemblage was dominated by Cladocerans. This indicates a marked seasonal change in the relative abundances of Zooplanktons.

Site	Season	Shannon diversity index $H = \sum - (P_i * \ln P_i)$	Simpsons diversity Index (D) $H = \sum (n_i/n)^2$	Evenness index $J = H/\ln S$
Control	Pre-monsoon	3.213	0.050	0.896
Control	Post-monsoon	3.20708	0.04442	0.46184
Experimental	Pre-monsoon	2.976	0.071	0.844
Experimental	Post-monsoon	3.11335	0.04924	0.47109

Table: 21 Phytoplankton diversity index during pre- and post- monsoon

The Shannon Diversity Index (H') showed a relatively high diversity of phytoplankton at both locations. At the control site, the Shannon index was 3.213 during the pre-monsoon period and showed a very slight decrease to 3.208 during the post-monsoon period. In contrast, the experimental site showed a rise from 2.976 to 3.11 during the pre-monsoon period. Phytoplankton diversity was thus relatively stable in the control site but increased significantly in the experimental site after monsoons.

The Simpson index (D) was 0.050 at the control site during the pre-monsoon period and slightly lower at 0.04442 during the post-monsoon period. In the experimental site, the value decreased from 0.071 to 0.049 between pre-monsoon and post-monsoon periods. As these values are representative of the Simpson dominance form ($D = (n-1) / (d^2)$), the lower values after the monsoon indicate a decrease in the dominance of individual phytoplankton groups and a relatively more heterogeneous community.

The Evenness index (J) was significantly higher at both locations during the pre-monsoon period. The control station recorded a balance value of 0.896, which dropped to 0.462 after the monsoon. Similarly, the experimental site showed a decrease of 0.844 to 0.471. This suggests that the relative abundance of phytoplankton taxa in the pre-monsoon period was more evenly distributed in accordance with the index values calculated.

There was a clear seasonal difference in the diversity indices of phytoplankton. The control site maintained almost the same Shannon diversity between pre- and post-monsoon seasons (3.213 and 3.208), while the experimental site increased Shannon diversity from 2.976 to 3.11. Simpson's dominance values decreased in both locations during the post-monsoon period, especially in the experimental site, suggesting a decreased dominance. However, the dispersion index decreased significantly in both locations during the post-monsoon period. In general, the phytoplankton community showed a relatively stable diversity at the control site and an increase in Shannon diversity at the experimental site after monsoon. Simpson's

dominance decreased in both locations, and calculated dispersion values were higher during the pre-monsoon period.

Site	Season	Shannon diversity index $H = \sum - (P_i * \ln P_i)$	Simpsons diversity Index (D) $H = \sum (n_i/n)^2$	Evenness index $J = H/\ln S$
Control	Pre-monsoon	2.436	0.104	0.878
Control	Post-monsoon	2.7156	0.066457	0.50222
Experimental	Pre-monsoon	1.809	0.222	0.823
Experimental	Post-monsoon	2.3645	0.10005	0.485

Table: 22 Zooplankton diversity index during pre-and post -monsoon

The Shannon Diversity Index (H') was higher in the control site in both the seasons. At the control site, Shannon diversity increased from 2.436 pre-monsoon to post-monsoon values of 2.716. Similarly, in the experimental site, the index increased from 1.809 pre-monsoon to post-monsoon season by 2.36. This suggests that zooplankton diversity increased in both pre- and post-monsoon in control as well as experimental samples. However, the Simpson Diversity Index (D) showed a decrease in both control and experimental samples during the post-monsoon period. The values decreased from 0.104 pre-monsoon to post-monsoon 0.0665 and from 0.222 to 0.100 at the control site. As the values presented are in the form of Simpson's dominance (D), the lower values after the monsoon indicate a lower dominance and a more varied distribution of zooplankton taxa at this time of the year.

The evenness index was higher in both control and experimental locations during the pre-monsoon period. The control site recorded a mean of 0.878 during the pre-monsoon period, which dropped to 0.502 during the post-monsoon period. Similarly, in the experimental site it decreased from 0.823 to 0.485. Thus, despite the increase in Shannon diversity after monsoon season, the calculated equilibrium values indicate that the distribution of taxa is less uniform after monsoon season.

Overall, the results show clear seasonal variations in the diversity of zooplankton. The Shannon diversity increased from pre- to post-monsoon in both control and experimental sites, with control sites consistently showing a higher Shannon diversity than experimental sites. The Simpson dominance index declined during the post-monsoon period, indicating a decrease in the dominance of individual groups. However, the balance index showed a significant decrease in both locations during the post-monsoon period, suggesting that the increase in diversity has not been accompanied by a more homogeneous distribution of individuals between the taxa recorded.

Source of Variation	SS	df	MS	F	P-value	F crit
Sample	16032.14	1	16032.14	65.13349	2.7E-08	4.259677
Columns	4577.286	1	4577.286	18.59605	0.000239	4.259677

Interaction	343	1	343	1.3935	0.249383	4.259677
Within	5907.429	24	246.1429			
Total	26859.86	27				

Table: 23 Two-Way ANOVA of Phytoplankton's Abundance across Sites and Seasons

The two-way ANOVA revealed significant differences in phytoplankton abundance between the control and experimental sites ($F = 65.13$, $p < 0.001$), with the control site consistently recording higher values. Seasonal variation was also statistically significant ($F = 18.59$, $p < 0.001$), indicating that phytoplankton abundance increased during the post-monsoon period, likely due to enhanced nutrient influx and hydrological changes associated with monsoonal dynamics. In contrast, the interaction between site and season was not significant ($F = 1.39$, $p = 0.249 > 0.05$), suggesting that seasonal effects were uniform across both sites. Collectively, these findings demonstrate that spatial and temporal factors independently influenced phytoplankton abundance, while their combined interaction did not, underscoring the role of localized environmental conditions and seasonal variability in shaping phytoplankton distribution.

Source of Variation	SS	df	MS	F	P-value	F crit
Sample	1560.036	1	1560.036	45.40644	5.71E-07	4.259677
Columns	211.75	1	211.75	6.163202	0.02043	4.259677
Interaction	15.75	1	15.75	0.45842	0.504838	4.259677
Within	824.5714	24	34.35714			
Total	2612.107	27				

Table: 24 Two-Way ANOVA of Zooplankton Abundance across Sites and Seasons

The two-way ANOVA analysis demonstrated that zooplankton abundance varied significantly between the control and experimental sites ($F = 45.41$, $p < 0.001$). The control site consistently exhibited higher mean values compared to the experimental site, indicating that spatial differences exerted a meaningful influence on zooplankton distribution. Seasonal variation was also statistically significant ($F = 6.16$, $p = 0.020 < 0.05$), with greater abundance observed during the post-monsoon period. This suggests that monsoon-driven hydrological changes and nutrient influx contributed to enhanced zooplankton growth following the monsoon.

In contrast, the interaction effect between site and season was not statistically significant ($F = 0.46$, $p = 0.505 > 0.05$), implying that seasonal influences were consistent across both sites and no combined site $\times$ season effect was detected.

Overall, these findings indicate that both spatial and temporal factors independently influenced zooplankton abundance, while their interaction did not. The higher abundance at the control site and during the post-monsoon period highlights the role of localized environmental conditions and seasonal variability in shaping zooplankton dynamics.

DISCUSSION

This study tried to examine any likely impact of Ash Pond Slurry discharge of Planktonic communities the seasonal dynamics of the plankton communities. The spread-over water of Hirakud Dam Reservoir as well as other small waterbodies close to ITPS, Banharpali as experimental site and Rampela as control site. The composition and abundance of plankton was assessed during the pre-monsoon and post-monsoon periods at seven sampling stations in each site. The study shows distinct spatial and seasonal variations in the phytoplankton and zooplankton communities as well as in the zooplankton population.

Seasonal variation in phytoplankton

Phytoplankton communities showed clear differences between the pre- and post-monsoon period. During the pre-monsoon period, chlorophyll was the major phytoplankton group in both control and experimental sites, accounting for 37.60 and 34.61 percent of the total. The second main group was Cyanophyceae, accounting for 33.62 percent in the control setting and 40.34 percent in the experimental samples. Bacillophyceae accounted for 13.56 and 13.00 percent of the total, while Euglenophyceae and Chrysophyceae accounted for a relatively smaller share. Vasculariaceae were only observed in the control site (0.55%)

During the post-monsoon period, the relative contribution of chlorophyll increased to 43.88 percent in the control and 40.61 percent in the experimental setting. Bacillariophyceae also increased significantly, with 34.91 and 34.62 percent. In contrast, cyanophyceae decreased to 13.79 percent in the control and 20.91 percent in the experimental samples. Euglenophyceae also decreased, while Chrysophyceae decreased in the control with no decrease in the experimental samples. Vaculariaceae were not recorded in either of these locations during the post-monsoon period

During the post-monsoon period, phytoplankton abundance increased across both locations. The control site recorded rise in average numbers from 129.57 Nos./ml in the pre-monsoon season to 148.14 Nos./ml in the post-monsoon season, while the experimental site increased from 74.71 to 107.29 Nos./ml.

A two-way ANOVA confirmed that these trends were statistically significant. Season was a key factor driving phytoplankton density ($F = 18.596$, $p = 0.000239$), and the overall abundance differed substantially between the control and experimental sites ($F = 65.133$, $p = 2.7 \times 10^{-8}$).

Because the interaction between site and season was not significant ($F = 1.394$, $p = 0.249$), both location and time of year affected phytoplankton levels independently, meaning the seasonal shifts occurred similarly at both sites.

Phytoplankton Community Dynamics and Diversity

Phytoplankton diversity metrics highlighted clear shifts between seasons. At the control site, Shannon diversity remained steady, shifting marginally from 3.213 before the monsoon to 3.208 after. Conversely, the experimental site experienced a rise from 2.976 to 3.113, demonstrating a more pronounced seasonal enhancement in species diversity.

Simpson's dominance index decreased post-monsoon at both locations—dropping from 0.050 to 0.044 at the control site and from 0.071 to 0.049 at the experimental site. Given that lower values reflect diminished

dominance by specific individual taxa, this drop points to a less to heavy phytoplankton community following the rains.

However, species evenness fell sharply post-monsoon, dropping from 0.896 to 0.462 at the control site and from 0.844 to 0.471 at the experimental site. This indicates that while overall diversity remained stable or increased, individuals were distributed far less uniformly across taxa.

Zooplankton Seasonal Restructuring and Composition

Zooplankton populations also underwent prominent seasonal shifts. Pre-monsoon sampling revealed Copepoda as the dominant taxon at the control site (10.57 Nos./ml), followed by Rhizopoda (7.71 Nos./ml). At the experimental site, Rhizopoda led (5.00 Nos./ml), with Copepoda secondary (3.29 Nos./ml).

Following the monsoon, Cladocera replaced these groups as the prevailing zooplankton at both locations. Its mean density grew from 4.71 to 12.00 Nos./mL at the control site and from 0.71 to 8.43 Nos./ml at the experimental site, whereas Copepoda abundance dropped to 7.71 and 3.14 Nos./ml, respectively. The post-monsoon emergence of Branchionide and Crustacea further underscores this structural turnover.

Relative abundance trends mirrored these density shifts. Pre-monsoon zooplanktons were primarily composed of Copepoda and Rhizopoda, but post-monsoon communities shifted heavily toward Cladocera. At the experimental site, Cladocera jumped from 6.10% to 45.04% of the community, while Rotifera disappeared entirely from 23.17%.

Zooplankton Diversity Metrics

Zooplankton diversity improved across both sites after the monsoon. Shannon diversity values climbed from 2.436 to 2.72 at the control site and from 1.809 to 2.3645 at the experimental site, confirming richer community diversity during the post-monsoon period.

Simpson's dominance index declined from 0.104 to 0.066 at the control site and from 0.222 to 0.100 at the experimental site, signaling reduced numerical dominance by single groups. However, species evenness fell from 0.878 to 0.502 at the control site and from 0.823 to 0.485 at the experimental site, showing that higher post-monsoon diversity was driven by taxa presence rather than an even balance of individuals among groups.

Statistical and Ecosystem Synthesis

A two-way ANOVA confirmed that location and season both significantly altered zooplankton abundance. Inter-site differences were highly significant ($F = 45.406$, $p = 5.71 \times 10^{-7}$), as was seasonal variation ($F = 6.163$, $p = 0.02043$). The interaction between site and season was not significant ($F = 0.458$, $p = 0.505$), establishing that seasonal gains occurred independently of location.

In summary, the planktonic ecosystem showed strong seasonal reorganization. Post-monsoon conditions boosted phytoplankton abundance through expanded Chlorophyceae and Bacillariophyceae populations shifting away from pre-monsoon Cyanophyceae prominence while zooplankton transitioned from Copepoda- and Rhizopoda-heavy communities to Cladocera dominated assemblages.

ACKNOWLEDGEMENTS

The authors are thankful to the Authorities of Sambalpur University and Head ,P.G.Department of Environmental Sciences of TERI University, Sambalpur University and Chairman, DEFT ,Bhubaneswar for providing administrative and academic support to carry out the project. Full financial support granted

by the Odisha Power Generation Corporation to undertake a comprehensive project on Social audit is gratefully acknowledged.

REFERENCES

1. Adriano, D.C., Page, A.L., Elseewi, A.A., Chang, A.C. and Straughan, I. 1980. Utilisation and disposal of fly ash and other coal residues in terrestrial ecosystems; a review. *Journal of Environmental Quality*, 9, 333-34. <https://doi.org/10.2134/jeq1980.00472425000300001x>
2. Bhat, N. A., Wanganeo, A., & Raina, R. (2015). Variability in water quality and phytoplankton community during dry and wet periods in the Tropical Wetland, Bhopal, India. *Journal of Ecosystem & Ecography*, 5(2), 1.
3. Borm, P.J. (1997). Toxicity and occupational health of Coal Fly Ash (CFA): A review of data and comparison to coal mine dust. *The Annals of Occupational Hygiene*, 41, 659-676. [http://dx.doi.org/10.1016/S0003-4878\(97\)00026-4](http://dx.doi.org/10.1016/S0003-4878(97)00026-4)
4. Ghio, A.J., Silbajoris, R., Carson, J.L. and Samet, J.M. (2002) Biological effect of Oil Fly ash. *Environmental Health Perspectives*, 110, 89-94. <https://doi.org/10.1289%2fehp.02110s1189>
5. Hossain, M. R. A., Pramanik, M. M. H., & Hasan, M. M. (2017). Diversity indices of plankton communities in the River Meghna of Bangladesh. *International Journal of Fisheries and Aquatic Studies*, 5(3), 330-334.
6. Jaunjal, A., & Gaupale, T. (2026). Impact of Coal Fly Ash on Freshwater Fishes and the Ecosystem. *International Journal of Ecology and Environmental Sciences*, 51(1), 37-43. <https://doi.org/10.55863/ijees.2025.0488>
7. Kakde, U. B., & Sitre, S. R. (2015). Effect of thermal power station effluents on wetland ecosystem. *International Journal of Researches in Biosciences, Agriculture and Technology*, 2(7), 252-259.
8. Mahananda, H. B., Mahananda, M. R., & Mohanty, B. P. (2005). Studies on the physico-chemical and biological parameters of a fresh water pond ecosystem as an indicator of water pollution. *Ecology environment and conservation*, 11(3/4), 537.
9. Manz, M., Weisflog, L., Kuhne, R. and Schurmann, G. (1999). Ecotoxicological hazard and risk assessment of heavy metal contents in agricultural soil of Central Germany. *Ecotoxicology Environmental Safety*, 42, 191-202. <https://dx.doi.org/10.1006/eesa.1998.1741>
10. Rankhamb, S. V., & Mulgir, V. S. (2024). Zooplankton Diversity as Bioindicators of Environmental Stress in Aquatic Ecosystems. *International Journal of Biological Innovations*, 6(1), 65-68.
11. Saha, D., Keshri, J. P., & Saha, N. C. (2020) Assessment of seasonal phytoplankton diversity of abandoned coal pits in Harabhanga village, Raniganj, West-Bengal with reference to pollution status caused by heavy metals. *International Journal of Ecology and Environmental Sciences*, 2(4), 59-66.
12. Spencer, D.F., Young, H.Y. and Greene, R.W. 1983. Alteration in Zooplankton community of a fly ash treated lake. *Hydrobiologia*, 107, 123-130. <https://doi.org/10.1007/BF00117427>
13. Walia, A. and Mehra, N. (1998). A seasonal assessment of the impact of coal fly ash disposal on the river Yamuna, Delhi II. *Biology. Water, Air and Soil Pollution*, 103, 315-339. <https://dx.doi.org/10.1023%2FA%3A1004953715779>
14. Yi, C., Yingjee, F., Yu, H., Xiaoling, L., Xu, W. and Zhang, T. (2024). A comprehensive review of toxicity of coal fly ash and its leachate in the ecosystem. *Ecotoxicology and Environmental Safety*, 269, 1-20. <https://doi.org/10.1016/j.ecoenv.2023.115905>